

Title

Directional transfer of chemical-gene effects from rodents to humans: an independence-guarded meta-analysis of the Comparative Toxicogenomics Database

Article type

Research Article

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Declaration of competing interests

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Data availability

The Comparative Toxicogenomics Database (<https://ctdbase.org>), NCBI `gene_orthologs` and Ensembl Compara are openly available. Analysis code and non-identifiable aggregate tables and figures: <https://github.com/january-msemakweli/Cross-Species-Directional-Transfer-Chemical-Gene-Effects> (Zenodo DOI to be added from the repository Release). No restricted or patient-level data were used.

CRedit authorship contribution statement

January G. Msemakweli: Conceptualization; Methodology; Software; Formal analysis; Data curation; Visualization; Writing (original draft); Writing (review and editing); Project administration. **Hussein Mohamed:** Methodology; Validation; Writing (review and editing). **Iddi Mapande:** Methodology; Validation; Writing (review and editing). **Oscar Punguti:** Data curation; Validation; Writing (review and

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Grammarly AI in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.